

Smart Restaurant Ordering & Kitchen Display System

Database Systems Engineering and Distributed Backend Development

Team 25 • Section 12 • CSIT

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PROJECT VIEW = 2

Introduction

- Traditional restaurant ordering mainly depends on communication between customers, waiters, and kitchen staff.
- Manual ordering can result in:
 1. Incorrect orders
 2. Delays in order processing
 3. Difficulty in tracking orders
 4. Challenges in managing restaurant operations
 5. Our project introduces a digital restaurant ordering and kitchen management system to make the process faster, simpler, and more efficient.

Abstract

Smart Restaurant Ordering & Kitchen Display System provides a digital solution for restaurant ordering and kitchen management.

- Customers scan a QR code at their table.
- They access the digital menu using their mobile device.
- Customers select food items, quantities, and add them to a cart.
- Orders are stored in the database and displayed on the kitchen dashboard.
- Kitchen staff can update orders through:

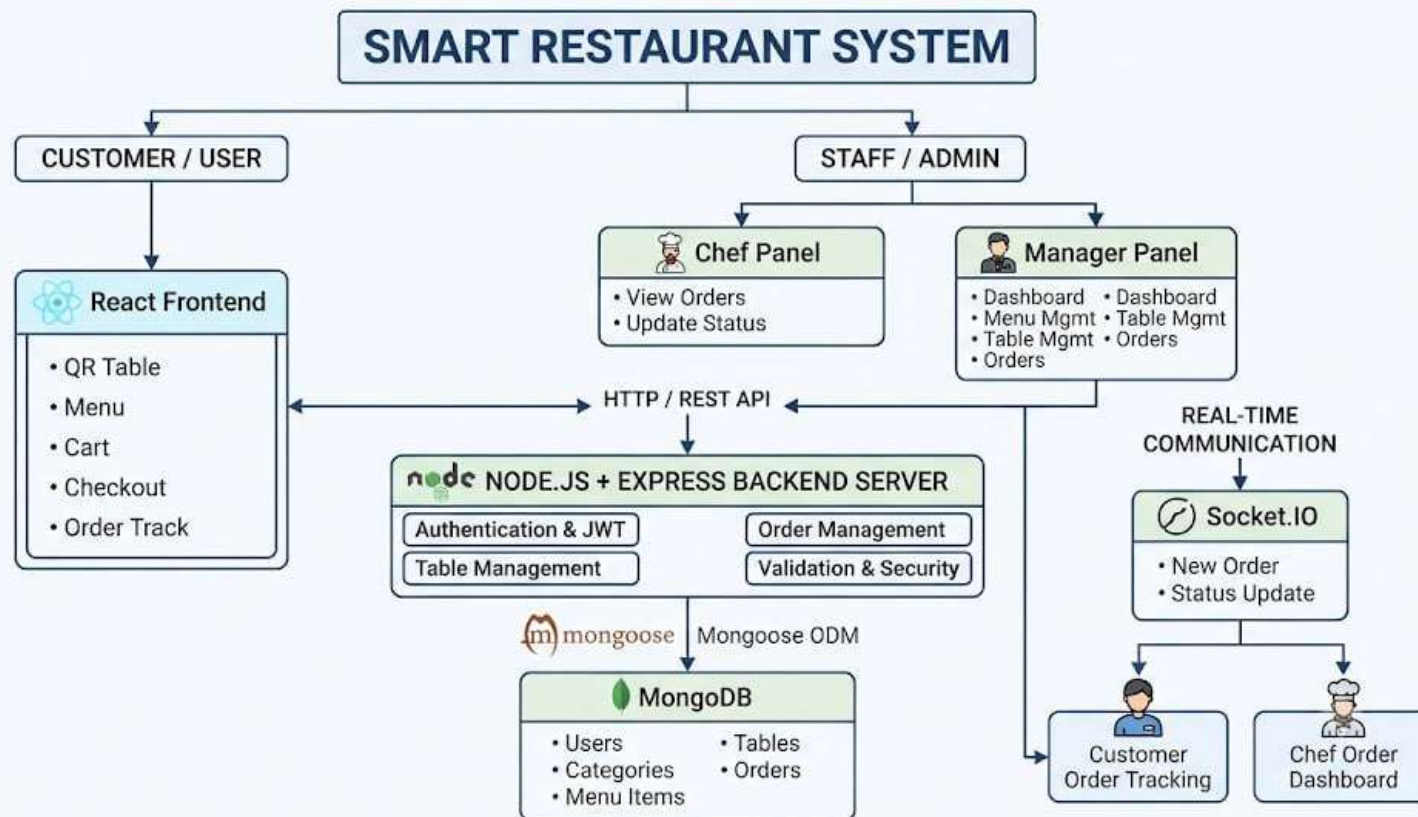
Pending → Preparing → Ready → Delivered

- Socket.io enables real-time order and status updates.

Objectives

- Reduce manual errors in restaurant ordering.
- Provide a simple QR-based ordering system.
- Enable customers to place orders directly.
- Store and manage restaurant data efficiently.
- Provide a real-time kitchen order dashboard.
- Improve communication between customers and kitchen staff.
- Automate bill calculation and invoice generation.
- Provide an efficient and user-friendly restaurant management solution

SYSTEM ARCHITECTURE



REST API

- **/api/auth** → Handles user registration, login, logout, and authentication.
- **/api/tables** → Handles restaurant table operations, such as viewing or managing tables.
- **/api/menu** → Handles food menu operations, such as viewing, adding, or updating menu items.
- **/api/orders** → Handles customer orders, such as creating, viewing, and updating orders.

```
55 // Mount Routers
56 app.use('/api/auth', authRoutes);
57 app.use('/api/tables', tableRoutes);
58 app.use('/api/menu', menuRoutes);
59 app.use('/api/orders', orderRoutes);
60
```

Admin Module

The admin module allows restaurant administrators to manage:

- Menu items
- Restaurant tables
- Customer orders
- Basic sales information
- Restaurant-related application data

The system also supports automatic bill calculation and printable invoices

Software / Technologies Used

React.js — Frontend

Express.js — REST API / Backend

Socket.io — Real-time communication

HTML5 & CSS3 — Web structure & styling

QR Code Generation — Table-based ordering

Antigravity — Listed tool/platform

Node.js — Backend

MongoDB — Database

JavaScript — Application logic

REST APIs — Client-server communication

Git / GitHub — Version control

Expected Outcomes

Minimize errors caused by manual order taking.

Reduce delays in order processing.

Improve order tracking and kitchen coordination.

Provide a convenient digital ordering experience.

Centralize restaurant data in a database.

Automate billing and invoice generation.

Improve overall restaurant management efficiency.

Conclusion

The Smart Restaurant Ordering & Kitchen Display System digitizes restaurant ordering and kitchen management using QR codes, a digital menu, database storage, and real-time communication.

The system aims to reduce manual errors, improve order processing, and provide an efficient, user-friendly digital restaurant management solution.